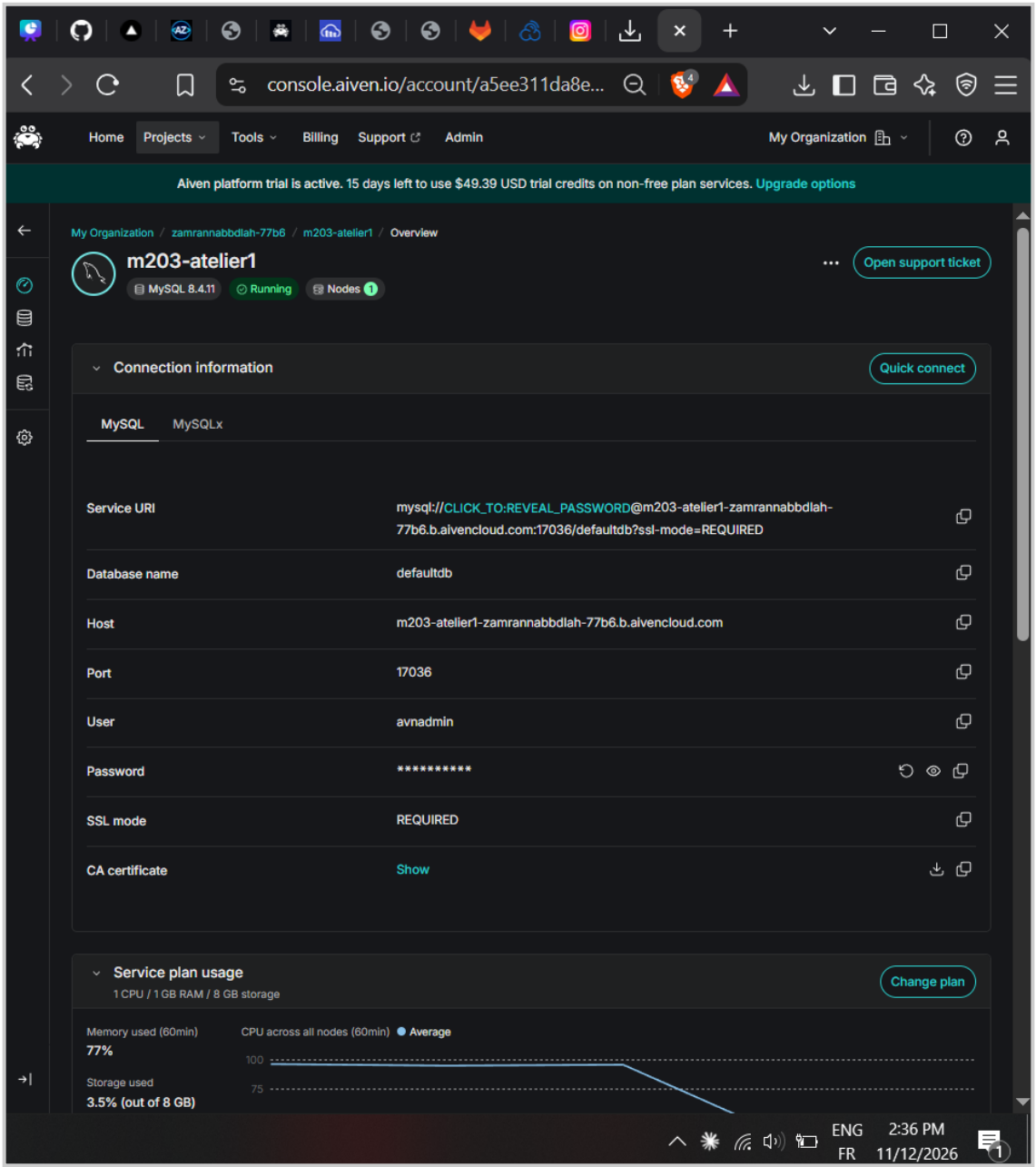
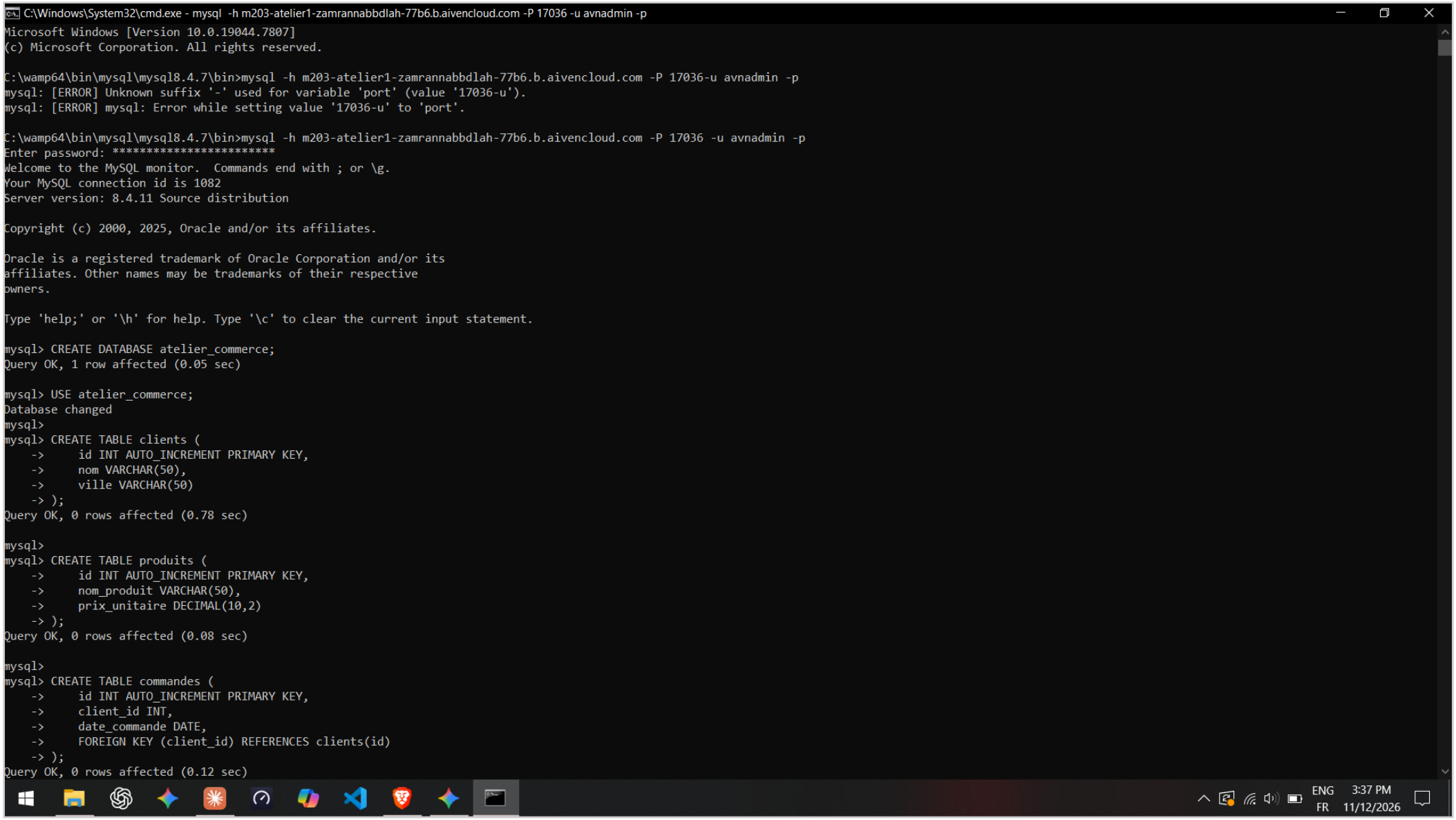
Module : M203 Gestion des données

Atelier 1: SQL Complet – Requêtes, Calculs, Optimisation et Sécurité




C:\Windows\System32\cmd.exe - mysql -h m203-atelier1-zamrannabbdlah-77b6.b.aivencloud.com -P 17036 -u avnadmin -p
Microsoft Windows [Version 10.0.19044.7807]
(c) Microsoft Corporation. All rights reserved.
C:\wamp64\bin\mysql\mysql8.4.7\bin>mysql -h m203-atelier1-zamrannabbdlah-77b6.b.aivencloud.com -P 17036-u avnadmin -p
mysql: [ERROR] Unknown suffix '-' used for variable 'port' (value '17036-u').
mysql: [ERROR] mysql: Error while setting value '17036-u' to 'port'.
C:\wamp64\bin\mysql\mysql8.4.7\bin>mysql -h m203-atelier1-zamrannabbdlah-77b6.b.aivencloud.com -P 17036 -u avnadmin -p
Enter password: **********
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 1082
Server version: 8.4.11 Source distribution
Copyright (c) 2000, 2025, Oracle and/or its affiliates.
Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
mysql> CREATE DATABASE atelier_commerce;
Query OK, 1 row affected (0.05 sec)
mysql> USE atelier_commerce;
Database changed
mysql>
mysql> CREATE TABLE clients (
-> id INT AUTO_INCREMENT PRIMARY KEY,
-> nom VARCHAR(50),
-> ville VARCHAR(50)
->);
Query OK, 0 rows affected (0.78 sec)
mysql>
mysql> CREATE TABLE produits (
-> id INT AUTO_INCREMENT PRIMARY KEY,
-> nom_produit VARCHAR(50),
-> prix_unitaire DECIMAL(10,2)
->);
Query OK, 0 rows affected (0.08 sec)
mysql>
mysql> CREATE TABLE commandes (
-> id INT AUTO_INCREMENT PRIMARY KEY,
-> client_id INT,
-> date_commande DATE,
-> FOREIGN KEY (client_id) REFERENCES clients(id)
->);
Query OK, 0 rows affected (0.12 sec)
ENG
FR
3:37 PM
11/12/2026

C:\Windows\System32\cmd.exe - mysql -h m203-atelier1-zamrannabbdlah-77b6.b.aivencloud.com -P 17036 -u avnadmin -p

```
Query OK, 0 rows affected (0.07 sec)

mysql> INSERT INTO clients (nom, ville)
    -> VALUES ('Alice', 'Paris'), ('Bob', 'Lyon'), ('Charlie', 'Toulouse'), ('Dina', 'Nice');
Query OK, 4 rows affected (0.08 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO produits (nom_produit, prix_unitaire)
    -> VALUES ('Ordinateur', 950), ('Souris', 20), ('Clavier', 45), ('Écran', 250);
Query OK, 4 rows affected (0.05 sec)
Records: 4  Duplicates: 0  Warnings: 0

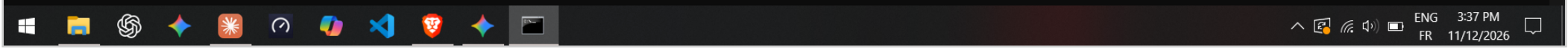
mysql>
mysql> INSERT INTO commandes (client_id, date_commande)
    -> VALUES (1, '2025-01-10'), (1, '2025-01-20'), (2, '2025-02-01'), (3, '2025-02-10');
Query OK, 4 rows affected (0.07 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO details_commandes (commande_id, produit_id, quantite)
    -> VALUES (1,1,1), (1,2,2), (2,3,1), (3,4,2), (4,2,3);
Query OK, 5 rows affected (0.05 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO paiements (commande_id, montant, mode_paiement, date_paiement)
    -> VALUES (1,990, 'Carte', '2025-01-12'), (2,45, 'Espèces', '2025-01-21'), (3,500, 'Virement', '2025-02-05');
Query OK, 3 rows affected (0.05 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SHOW TABLES;
+----------------------------+
| Tables_in_atelier_commerce |
+----------------------------+
| clients                    |
| commandes                  |
| details_commandes          |
| paiements                  |
| produits                   |
+----------------------------+
5 rows in set (0.09 sec)

mysql>
```



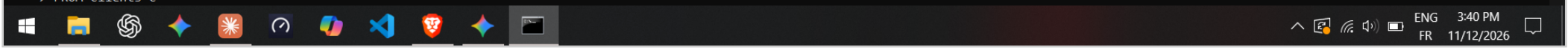
C:\Windows\System32\cmd.exe - mysql -h m203-atelier1-zamrannabbdlah-77b6.b.aivencloud.com -P 17036 -u avnadmin -p

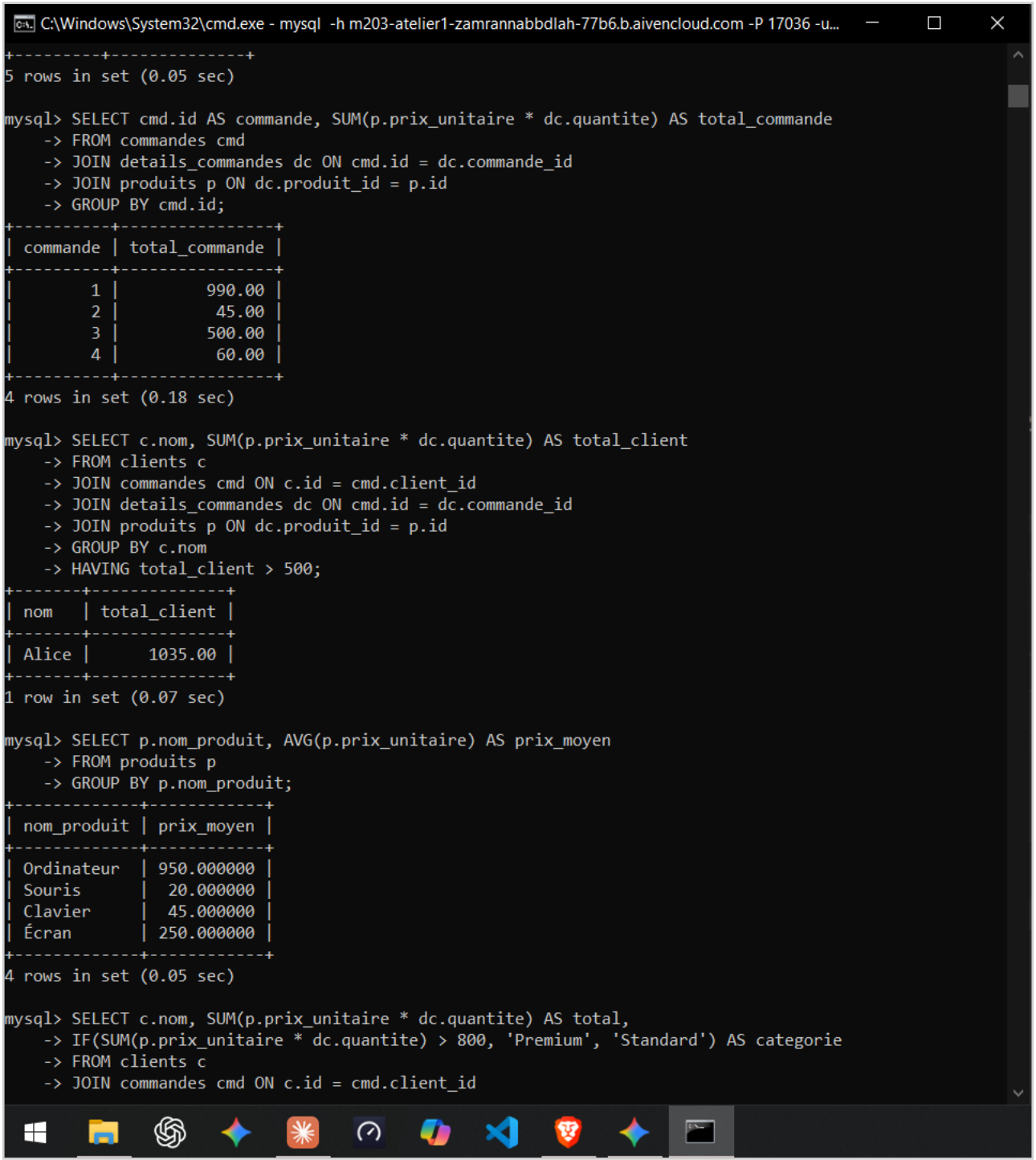
```
mysql> SELECT c.nom, cmd.id AS commande, p.nom_produit, dc.quantite
    -> FROM clients c
    -> INNER JOIN commandes cmd ON c.id = cmd.client_id
    -> INNER JOIN details_commandes dc ON cmd.id = dc.commande_id
    -> INNER JOIN produits p ON dc.produit_id = p.id;
+---------+----------+-------------+----------+
| nom     | commande | nom_produit | quantite |
+---------+----------+-------------+----------+
| Alice   |        1 | Ordinateur  |        1 |
| Alice   |        1 | Souris      |        2 |
| Alice   |        2 | Clavier     |        1 |
| Bob     |        3 | Écran       |        2 |
| Charlie |        4 | Souris      |        3 |
+---------+----------+-------------+----------+
5 rows in set (0.05 sec)

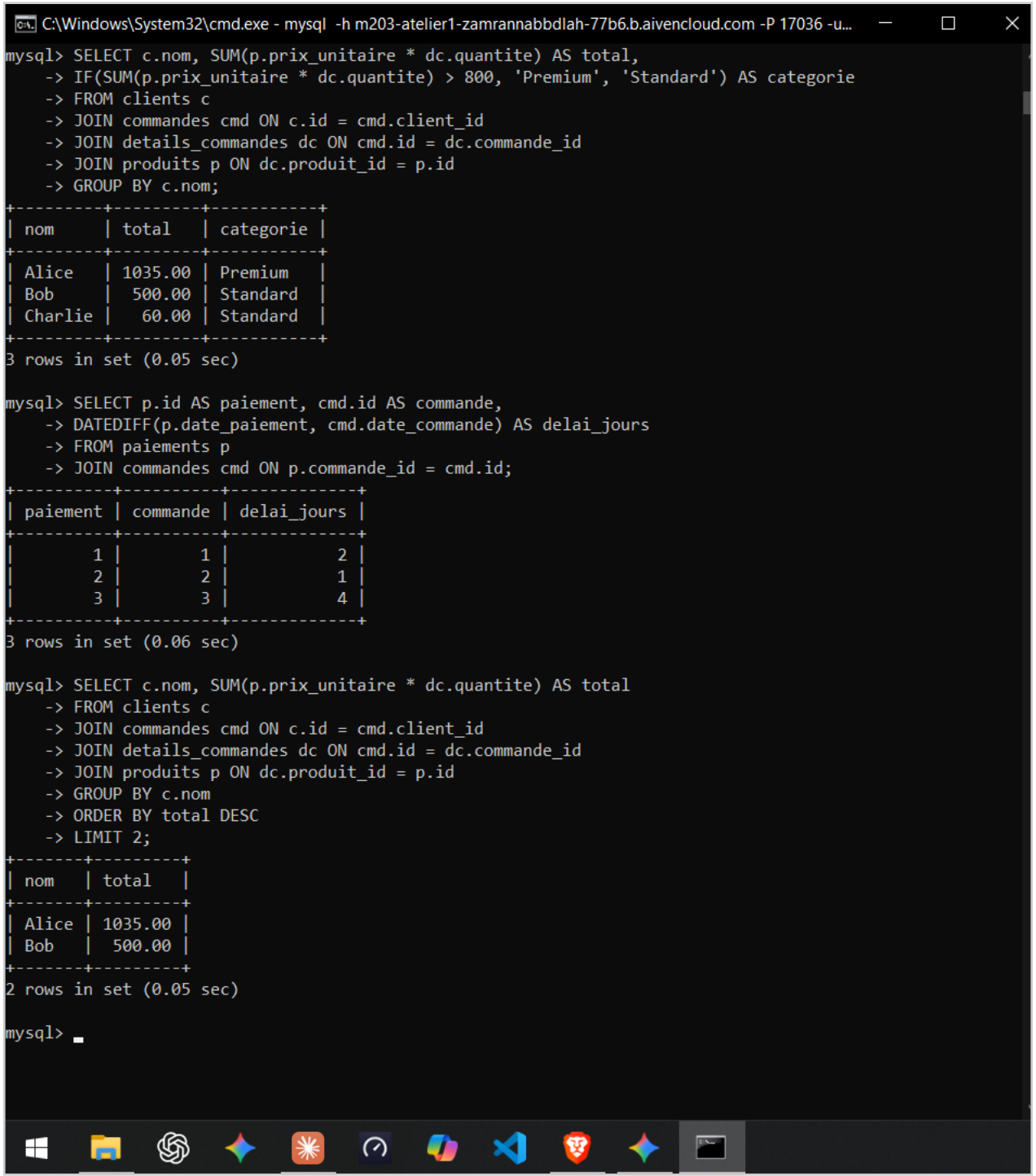
mysql> SELECT c.nom, cmd.id AS num_commande
    -> FROM clients c
    -> LEFT JOIN commandes cmd ON c.id = cmd.client_id;
+---------+--------------+
| nom     | num_commande |
+---------+--------------+
| Alice   |            1 |
| Alice   |            2 |
| Bob     |            3 |
| Charlie |            4 |
| Dina    |         NULL |
+---------+--------------+
5 rows in set (0.05 sec)

mysql> SELECT c.nom, cmd.id, cmd.date_commande
    -> FROM clients c
    -> RIGHT JOIN commandes cmd ON c.id = cmd.client_id;
+---------+----+---------------+
| nom     | id | date_commande |
+---------+----+---------------+
| Alice   |  1 | 2025-01-10    |
| Alice   |  2 | 2025-01-20    |
| Bob     |  3 | 2025-02-01    |
| Charlie |  4 | 2025-02-10    |
+---------+----+---------------+
4 rows in set (0.05 sec)

mysql>
mysql> SELECT c.nom, cmd.id AS num_commande
    -> FROM clients c
    -> LEFT JOIN commandes cmd ON c.id = cmd.client_id
    -> UNION
    -> SELECT c.nom, cmd.id AS num_commande
    -> FROM clients c
```




C:\Windows\System32\cmd.exe - mysql -h m203-atelier1-zamrannabbdlah-77b6.b.aivencloud.com -P 17036 -u...
+---------+--------------+
5 rows in set (0.05 sec)
mysql> SELECT cmd.id AS commande, SUM(p.prix_unitaire * dc.quantite) AS total_commande
-> FROM commandes cmd
-> JOIN details_commandes dc ON cmd.id = dc.commande_id
-> JOIN produits p ON dc.produit_id = p.id
-> GROUP BY cmd.id;
+----------+----------------+
| commande | total_commande |
+----------+----------------+
1	990.00
2	45.00
3	500.00
4	60.00
+----------+----------------+
4 rows in set (0.18 sec)
mysql> SELECT c.nom, SUM(p.prix_unitaire * dc.quantite) AS total_client
-> FROM clients c
-> JOIN commandes cmd ON c.id = cmd.client_id
-> JOIN details_commandes dc ON cmd.id = dc.commande_id
-> JOIN produits p ON dc.produit_id = p.id
-> GROUP BY c.nom
-> HAVING total_client > 500;
+-------+--------------+
| nom | total_client |
+-------+--------------+
| Alice | 1035.00 |
+-------+--------------+
1 row in set (0.07 sec)
mysql> SELECT p.nom_produit, AVG(p.prix_unitaire) AS prix_moyen
-> FROM produits p
-> GROUP BY p.nom_produit;
+-------------+------------+
| nom_produit | prix_moyen |
+-------------+------------+
Ordinateur	950.000000
Souris	20.000000
Clavier	45.000000
Écran	250.000000
+-------------+------------+
4 rows in set (0.05 sec)
mysql> SELECT c.nom, SUM(p.prix_unitaire * dc.quantite) AS total,
-> IF(SUM(p.prix_unitaire * dc.quantite) > 800, 'Premium', 'Standard') AS categorie
-> FROM clients c
-> JOIN commandes cmd ON c.id = cmd.client_id


C:\Windows\System32\cmd.exe - mysql -h m203-atelier1-zamrannabbdlah-77b6.b.aivencloud.com -P 17036 -u...
mysql> SELECT c.nom, SUM(p.prix_unitaire * dc.quantite) AS total,
-> IF(SUM(p.prix_unitaire * dc.quantite) > 800, 'Premium', 'Standard') AS categorie
-> FROM clients c
-> JOIN commandes cmd ON c.id = cmd.client_id
-> JOIN details_commandes dc ON cmd.id = dc.commande_id
-> JOIN produits p ON dc.produit_id = p.id
-> GROUP BY c.nom;
nom	total	categorie
Alice	1035.00	Premium
Bob	500.00	Standard
Charlie	60.00	Standard
3 rows in set (0.05 sec)
mysql> SELECT p.id AS paiement, cmd.id AS commande,
-> DATEDIFF(p.date_paiement, cmd.date_commande) AS delai_jours
-> FROM paiements p
-> JOIN commandes cmd ON p.commande_id = cmd.id;
paiement	commande	delai_jours
1	1	2
2	2	1
3	3	4
3 rows in set (0.06 sec)
mysql> SELECT c.nom, SUM(p.prix_unitaire * dc.quantite) AS total
-> FROM clients c
-> JOIN commandes cmd ON c.id = cmd.client_id
-> JOIN details_commandes dc ON cmd.id = dc.commande_id
-> JOIN produits p ON dc.produit_id = p.id
-> GROUP BY c.nom
-> ORDER BY total DESC
-> LIMIT 2;
nom	total
Alice	1035.00
Bob	500.00
2 rows in set (0.05 sec)
mysql>